School of Future Tech

Case Study Report

on

SkillSwap: Peer-to-Peer Learning

by

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1. Introduction to the Case Study

SkillSwap is a peer-to-peer learning model where a user can exchange their skills with someone else instead of using money to learn a skill. In a skill swap model, a user can teach a skill he or she knows and learn another skill from someone else. For instance, a user can teach Python programming and learn Spanish from someone else.

The skill swap model is based on collaborative learning where students and professionals can help each other in their skill development process.

2. Problem Statement / Case Background (Abstract)

Conventional learning systems are mostly dependent on paid learning materials, and this has made it difficult for users to acquire skills because they are not able to afford paid learning materials. There are many people out there with valuable skills, but they don't have a place to exchange their skills with others.SkillSwap has come up with a solution to this problem by providing a skill exchange platform. Instead of using money, users earn and spend credits by teaching others.Transparency is maintained on this platform by providing ratings, reviews, and mentor badges.

Abstract

SkillSwap is a digital platform developed for the purpose of facilitating the learning of skills in a more accessible and affordable way. The traditional method of learning new skills often involves the use of paid courses and resources, thereby limiting the ability of individuals to learn new skills due to financial constraints. On the contrary, there are individuals who have the necessary skills but are unable to share these skills with others. The development of the SkillSwap platform was motivated by the need for a skill exchange platform whereby individuals can teach others a skill in return for a credit, and the credit can be used to learn a new skill from another person instead of paying money.

3. Problem Statement / Case Study Design

The SkillSwap system is designed as a web-based application using HTML, CSS, and JavaScript.

The platform consists of three main sections:

1. Skills Page (skills.html)

This page displays different skill categories such as programming, languages, design, and marketing. Users can explore skills and filter them according to their interests.

2. Profile Page (profile.html)

The profile page shows user details including:

- Profile picture
- Skills offered
- Skills requested
- Mentor badges
- Credit balance

This helps users identify the right person to exchange skills with.

3. Reviews Page (reviews.html)

After completing a skill exchange session, users can rate each other and leave feedback. This improves trust and helps maintain quality in the learning community.

4. Methods & Algorithms Technology Applied

The platform uses the following technologies to implement the application:

HTML

HTML is used to develop the basic structure of the web pages such as:

- Skill categories
- User profile layout
- Review section

CSS

CSS is used to develop the user interface of the platform to make it look attractive to the user.

CSS is used to develop the following features:

- Profile badges
- Skill cards
- Layout styling

JavaScript

JavaScript is used to develop the application.

The features of the JavaScript application are as follows:

1. **Rating System Algorithm**

Users can rate the mentor from 1 to 5 stars. The application stores the ratings and calculates the average ratings displayed on the profile.

2. **Credit Balance Update**

When the user teaches the skill, the user earns credits. The user spends the credits when the user learns the skill.

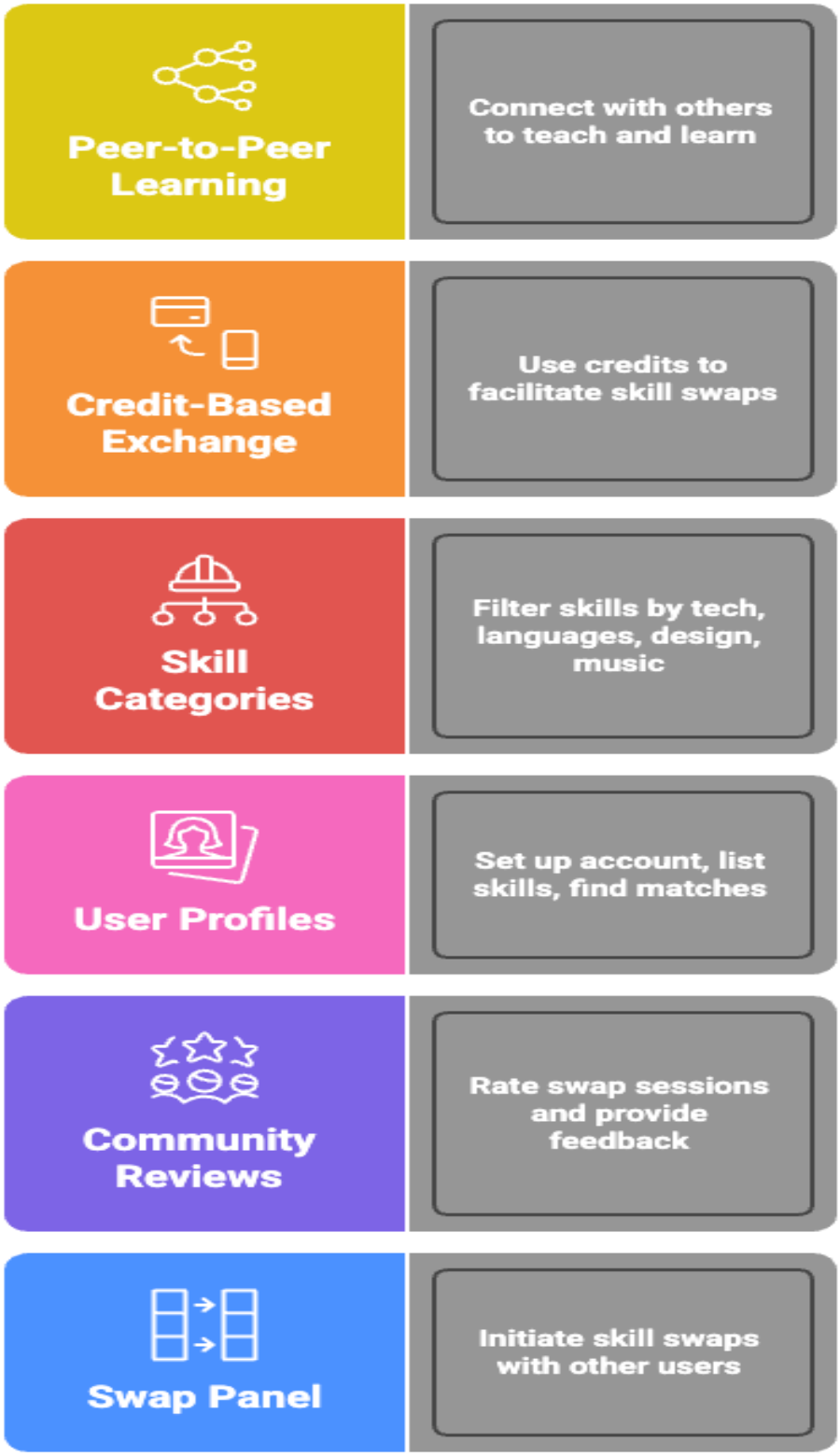
3. Skill Category Filter

JavaScript filters skills based on categories selected by the user.

4. Swap Request Modal

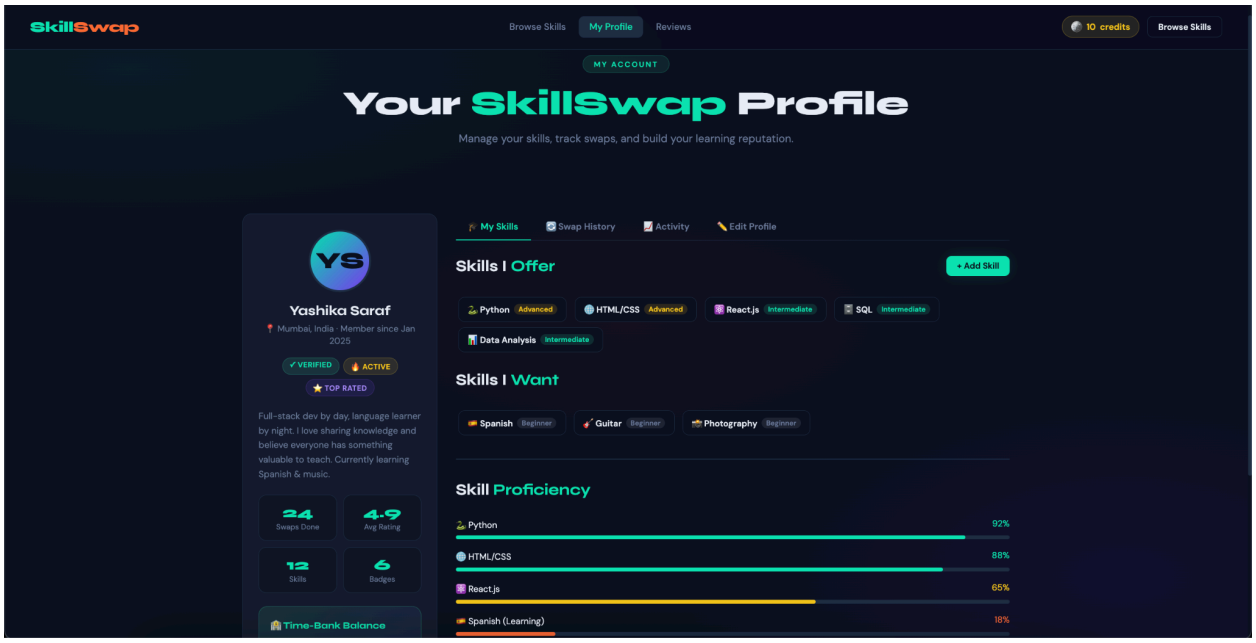
When a user wants to exchange skills, a pop-up modal appears where they can send a swap request.

Algorithm

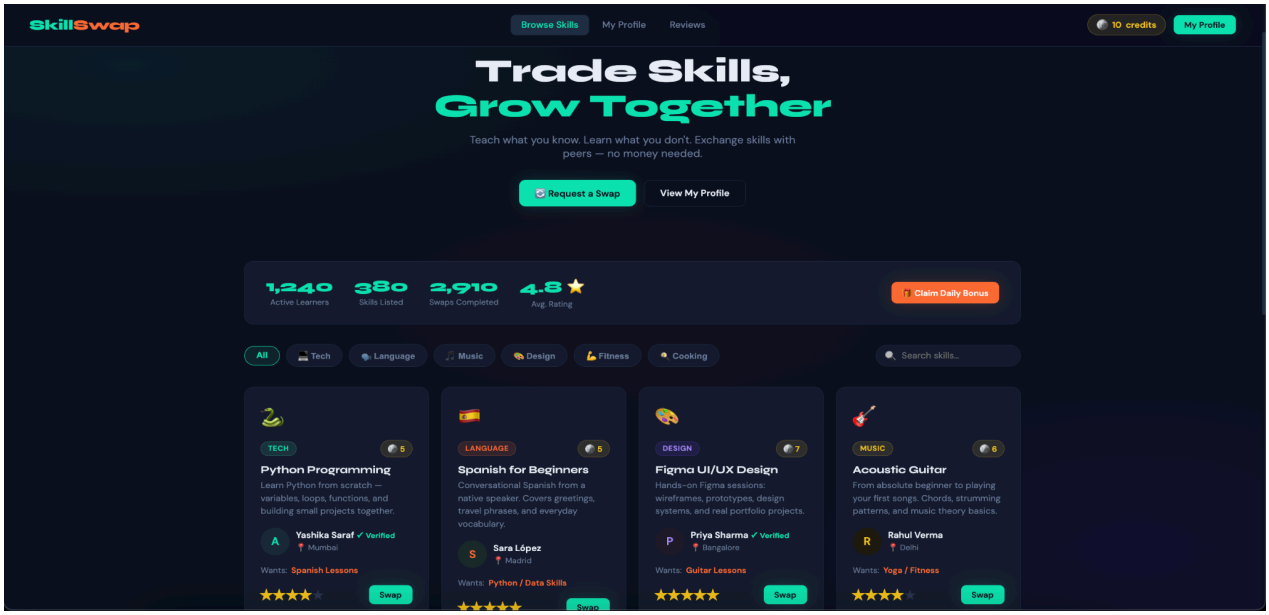

Peer-to-Peer Learning
Connect with others to teach and learn
Credit-Based Exchange
Use credits to facilitate skill swaps
Skill Categories
Filter skills by tech, languages, design, music
User Profiles
Set up account, list skills, find matches
Community Reviews
Rate swap sessions and provide feedback
Swap Panel
Initiate skill swaps with other users

5. Problem Statement / Case Study Implementation Details and Snapshots

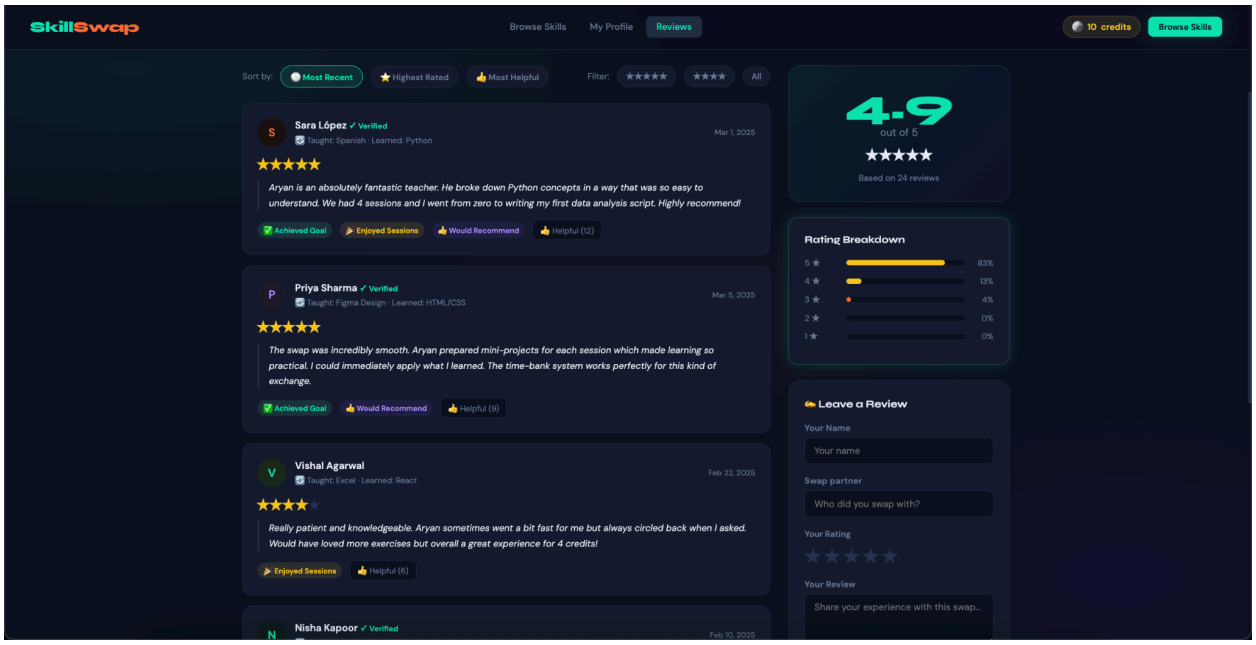
Output



This is the Profile section where the user can see their skills or credit points or what skills they have learnt.



This is the <u>Skills section</u> where all types skills are available and through credit points the user can swap with others and the user can also claim the daily bonus for the same



This is the <u>Reviews section</u> where all the users can just write their own opinion about this skillSwap or can just rate us.

6. Problem Statement / Case Study Results and Conclusion

Results

The development of the SkillSwap platform was successful, and it was created using HTML, CSS, and JavaScript. The system enabled users to explore different categories of skills, view user profiles, and request skills. The rating system was effective, as it enabled users to provide feedback after each session of skill swapping. The credit balance was effective, as it ensured fairness in the exchange of skills by adding credit when a user taught a skill and subtracting credit when they learned a skill.

Conclusion

The SkillSwap application has demonstrated that it is possible to implement a peer-to-peer learning system using a simple web-based application. The application has provided a platform where users can exchange skills rather than

using money, thus promoting learning within a community. The application has demonstrated how simple web programming techniques, such as HTML, CSS, and JavaScript, can be used to create a learning application. In the future, the application can be enhanced to incorporate user authentication, live chats, and video learning sessions.

7. References

- HTML CSS JS Programming Language Documentation
- ITM Skills University HTML CSS JS Case Study Material